

Course Syllabus Part I

CS 430 Software Engineering

3 credit hours

Course Description

This course introduces software engineering techniques that ensure development of well-designed, reliable, flexible, modular, and verified software and software systems. Development steps are explained, including software planning, specifications, coding, testing, and maintenance. Additional topics include software product development, cloud-based software, microservices architecture, code management and review, agile development, and DevOps.

Course Prerequisites

None

Course Objectives

Students who successfully complete the course should be able to:

1. Discuss the differences between functional and non-functional software requirements
2. Build graphical models to represent software systems
3. Defend the use of architectural design patterns
4. Compare software testing methodologies
5. Explain software reuse
6. Justify the benefits of continuous integration and system building

Grading Scale

<u>Letter Grade</u>	<u>Percentage Grade</u>	<u>Letter Grade</u>	<u>Percentage Grade</u>
A	≥ 92.5%	C	< 76.5% and ≥ 72.5%
A-	< 92.5% and ≥ 89.5%	C-	< 72.5% and ≥ 69.5%
B+	< 89.5% and ≥ 86.5%	D+	< 69.5% and ≥ 66.5%
B	< 86.5% and ≥ 82.5%	D	< 66.5% and ≥ 62.5%
B-	< 82.5% and ≥ 79.5%	D-	< 62.5% and ≥ 59.5%
C+	< 79.5% and ≥ 76.5%	F	< 59.5%

Topic Outline

- I. Software Engineering
 - A. Agile Software Development
 - B. Requirements Engineering
 - C. System Modeling
 - D. Architecture Design
 - E. Design and Implementation
 - F. Software Testing
- II. Architecture
 - A. Software Reuse
 - B. Component-Based Software Engineering
 - C. Distributed Software Engineering
 - D. Service-Oriented Software Engineering
 - E. Cloud Computing
- III. Software Management
 - A. Quality Management
 - B. Configuration Management

This syllabi update reflects grading scale policy updates effective 4/1/2024.